napping. To the top of your car you attach a series of firecrackers, one over the front of the hood, one over the rear of the trunk, and many in between; see Figure 1.3a. You set the firecrackers to detonate simultaneously as seen by you, just as you are passing my police station.

Figure 1.3b depicts this from your own viewpoint. Drawn vertically is the flow of time, as measured by you ("your time"). Drawn horizontally is distance along your car, from back to front, as measured by you ("your space"). Since the firecrackers are all at rest in your space (that is, as seen by you), with the passage of your time they all remain at the same horizontal locations in the diagram. The dashed lines, one for each firecracker, depict this. They extend vertically upward in the diagram, indicating no rightward or leftward motion in space whatsoever as time passes—and they then terminate abruptly at the moment the firecrackers detonate. The detonation events are depicted by asterisks.

This figure is called a *spacetime diagram* because it plots space horizontally and time vertically; the dashed lines are called *world lines* because they show where in the world the firecrackers travel as time passes. We shall make extensive use of spacetime diagrams and world lines later in this book.

If one moves horizontally in the diagram (Figure 1.3b), one is moving through space at a fixed moment of your time. Correspondingly, it is convenient to think of each horizontal line in the diagram as depicting space, as seen by you ("your space"), at a specific moment of your time. For example, the dotted horizontal line is your space at the moment of firecracker detonation. As one moves vertically upward in the diagram, one is moving through time at a fixed location in your space. Correspondingly, it is convenient to think of each vertical line in the spacetime diagram (for example, each firecracker world line) as depicting the flow of your time at a specific location in your space.